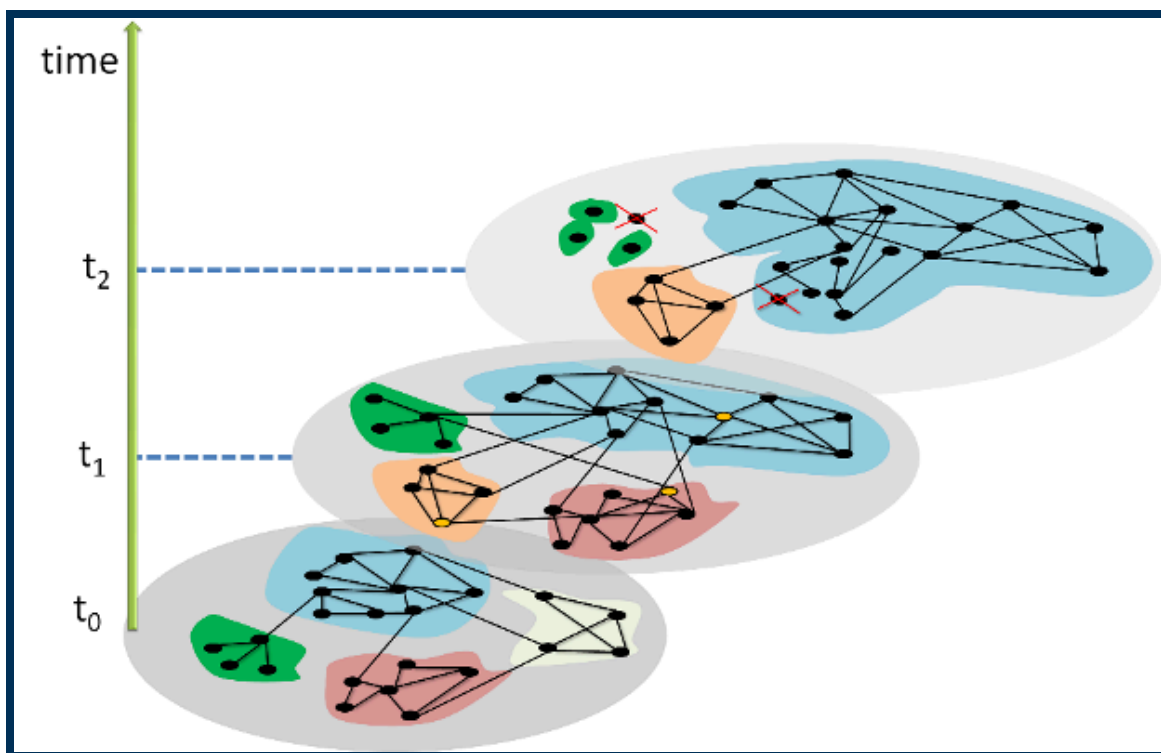


L1: Dynamics of Networks

An important component of Network Science is the focus on Dynamic Networks – systems that change over time through natural evolution, growth or other dynamic rewiring processes.

For example, the brain's neural network is changing dramatically during adolescence – but more recent research in neuroscience shows that brain connections also change when people learn something new or even when they meditate.



The image above shows how the community structure of a network may be changing over time. (Image Source: [The University of Florida](http://www.cise.ufl.edu/~tdinh/) [\(http://www.cise.ufl.edu/~tdinh/\)](http://www.cise.ufl.edu/~tdinh/))

Note that the white and red communities are gradually absorbed by the blue and the green community gradually collapses.

We will study algorithms that can detect and quantify such dynamic processes in networks.

Another important problem in Network Science is the study of Dynamic Processes on Networks. Here, the network structure remains the same – but there is a dynamic

process that is gradually unfolding on that network.

For example, the process may be an epidemic that spreads through an underlying social network.

For certain viruses, such as HIV, the state of each human can be one of the following: healthy but susceptible to the virus, infected by the virus but not yet sick, or sick (*symptomatic*).

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The **video above** shows a simulation of the spread of the H1N1 virus over the global air transportation network. The H1N1 outbreak started in Mexico in 2009 and it quickly spread throughout the world mostly through air transportation.

An important question in Network Science: how does the structure of the underlying network affect the spread of such epidemics?

As we will see later in the course, certain network properties enhance the spread of epidemics to the point that they can become pandemics before any intervention is

possible. The only way to prevent such pandemics is through immunizations when they are available.